

Fiche d'auto-évaluation 05

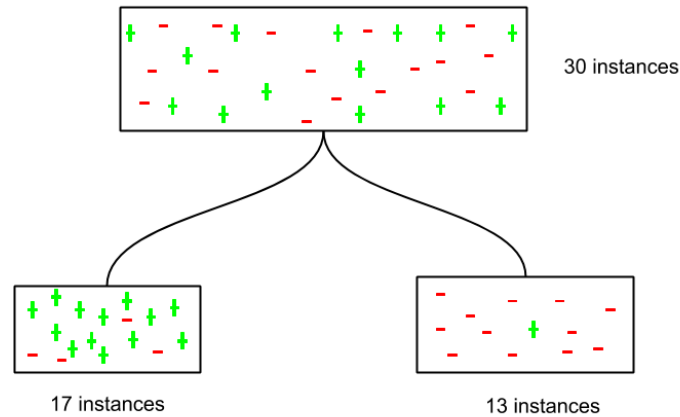
June 24, 2026

1. Consider training a neural network. Draw an example plot of training and validation loss vs. epochs in which training ended too soon (it would have been better to continue with more epochs).
2. Draw a plot like the previous, assuming now that your resulting model overfits the training dataset.
3. Draw a plot like the previous, assuming now that your resulting model underfits the training dataset
4. Are there cases, in Machine Learning, when gradient descent is guaranteed to converge?
5. Tell if the following way of applying scaling (Standard scaling) is correct or not. If not, correct it.
 - (a) Partition the dataset $\mathcal{D}$ in training dataset $\mathcal{D}^{\text{train}}$ and test dataset $\mathcal{D}^{\text{test}}$.
 - (b) Put the test set aside.
 - (c) Compute the standard deviation σ_j^{train} and the average μ_j^{train} of each column j on the training set.
 - (d) Transform all original values $x_j^{(i)}$ in $\frac{x_j^{(i)} - \dots}{\dots}$ (Complete the formula).
 - (e) Train a Machine Learning Model on the transformed training set.
 - (f) Take the test set $\mathcal{D}^{\text{test}}$, compute the minimum value $\min_j^{\text{test}}$, the standard deviation σ_j^{test} and the average μ_j^{test} of each column j on the test set $\mathcal{D}^{\text{test}}$.
 - (g) Transform $\mathcal{D}^{\text{test}}$ with a similar formula as before, now using $\sigma_j^{\text{test}}, \mu_j^{\text{test}}$
 - (h) Use the trained model to make predictions on the transformed test set.
6. If we apply 7-fold cross validation, how many models do we need to train?
7. You want to do Model selection: you want to use a type model, say Polynomial Regressions, and you want to select the best hyper-parameters (give some example of hyper-parameter). How can you use grid search and

cross-validation together? Explain the correct procedure, step by step. (this procedure is implemented in `sklearn-model_selection.GridSearchCV`).

8. When using Random Forests, are there cases where, increasing the number of trees, the accuracy of the forest decreases? If yes, which case?
9. What is the difference between random forest, bagging trees and extra-trees?
10. If you use a set of trees to do your prediction, is it better they are all similar or is it better they are different?
11. What is the CART algorithm used for? Describe it step by step. Then, explain how CART algorithm is integrated in a random forest, writing a pseudo-code.
12. If we increase the number of trees in a random forest, does the stability of the model faced with data perturbations increase or decrease?
13. Is a decision tree a linear or non-linear classifier? And a random forest?
14. In which sense a random forest is interpretable? (Give at least two arguments)
15. How can you compute the feature importance based on an ensemble of trees?
16. Is it necessary or at least useful to scale the dataset before using a decision tree or an ensemble of trees?
17. What does “linear classifier” mean?
18. Write the formula of the Gini impurity index and of the entropy with three classes.
19. Write the formula of the information gain.
20. Consider the following split¹

¹Picture from <https://www.hackerearth.com/practice/machine-learning/machine-learning-algorithms/ml-decision-tree/tutorial/>



Compute the information gain of the split, considering entropy and Gini impurity index.

21. Can information gain be negative? If yes, in which case?
22. What is the range of values that Gini index can take, when we have $K=2$ classes? And when $K=3$? And for any generic K ?
23. Answer the previous questions, considering now the entropy.
24. Is it good to have a fully grown tree? why?
25. Why is pruning important? For training or testing? What are the criteria that can be used to prune a tree?
26. How is randomness introduced in Bagging, Random Forests and Extra-trees?
27. Is the concept of bagging only limited to tree-based learning? Or can it also be applied with other types of models?
28. Suppose your classification model is an ensemble of decision trees. If we increase the diversity between the trees, is model stability faced with data perturbations increasing or decreasing? And if we increase the number of trees?
29. Explain how to train a random forest for regression.
30. Write the formula of the feature importance.
31. Suppose you train a random forest, and you try different configuration adding more and more trees. Is there a point when it is better to stop? When?